

User-Defined Gestures for Surface Computing

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01 BACKGROUND

2008: Where Touch Stood

1

1985

Buxton's multi-touch tablet, Toronto. Multi-touch research begins in academia.

2

2001–2006

DiamondTouch (MERL), SmartSkin (Sony), Jeff Han's viral TED demo in Feb 2006.

3

2007

iPhone announced Jan; on sale June

Microsoft Surface unveiled May

4

Aug 2008

The study we all read for the homework assignment. Patent filed 4 Aug 2008. Surface in AT&T stores since April. Paper presented in 2009

5

2009

CHI Boston, April. Windows 7 with multi-touch, October

Ordinary adults already had gesture-capable devices in hand — laptop touchpads (1994), Palm Graffiti (1996), iPod click wheel (2002), Nintendo DS (2004), Wii (2006) yet no coherent, user-derived gesture vocabulary existed for tabletop surfaces.



YouTube

Mitsubishi DiamondTouch Warcraft 3 demo

Tech demo using WarCraft 3 DiamondTouch is front-projected and uses an array of antennas embedded in the touch surface. Each antenna transmits a unique signal. Each user has a separate

01 BACKGROUND

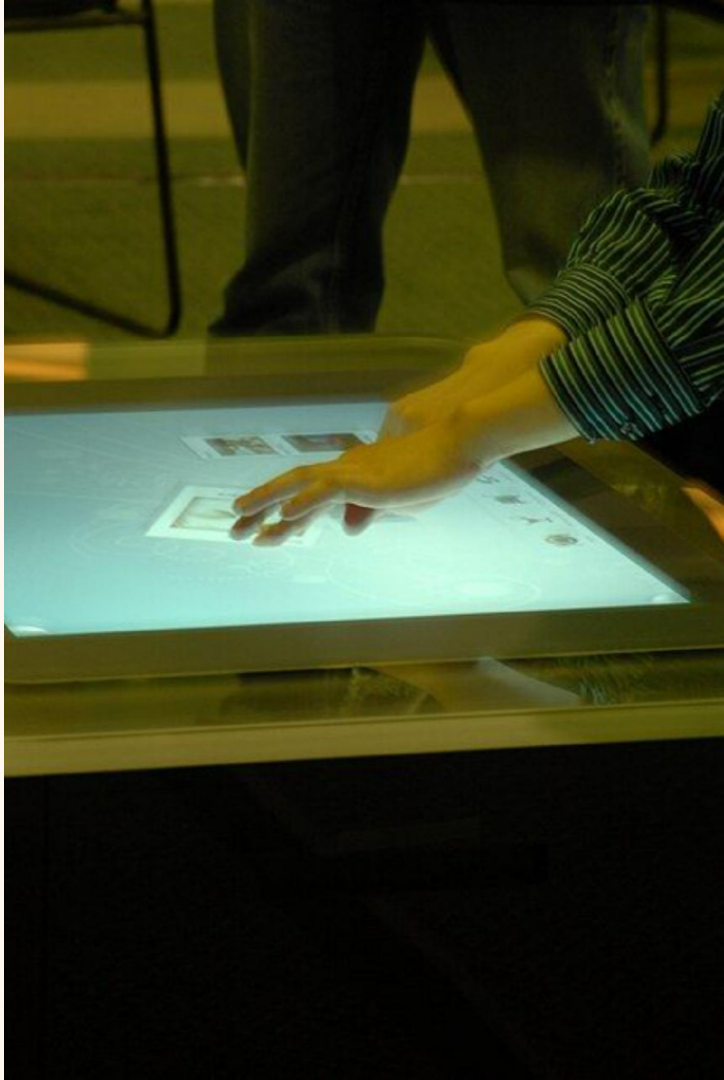
Why this Study, and what it was:

Why

- Tabletops for individuals, groups, multi-display rooms, and fun
- Keyboard and mouse "no longer preferable"
- Gestures "defined by system designers"
- Public tables demand "immediate usability"

What they did

- 20 non-technical adults at a Surface prototype
- 27 commands: see the effect, perform the cause; 1 and 2 hands; think aloud; no recognizer
- 1,080 gestures collected and logged
- Output: a taxonomy, agreement scores, and 48 gestures for 22 commands



02 THE PROBLEM

Gestures Built for recognition

> Gestures "defined by system designers," "chosen out of concern for reliable recognition"

Built from the Recognizer Outward

Prior tabletop systems were never designed with user input. Wu's tabletop gestures were "not elicited from users" (p. 1085). Malik et al. constrained inputs to 1, 2, 3, or 5 fingers — an approach the authors flag as "may be problematic" (p. 1085). In principle, "almost anything one can do with one's hands" could be a gesture (p. 1083), yet designers kept picking what the recognizer could handle reliably.

Users Had Been Asked Before

Good et al. (1984), Epps (2006), and Nielsen (2004) all elicited user input for specific commands (p. 1084). The authors claim to be "first to employ users, rather than principles" to construct a *complete* surface gesture set (p. 1084). The gap was not asking users at all — it was that nobody had built an entire vocabulary from their agreement.

The core problem: Users had been asked before. Nobody had built a whole surface gesture set from the users concensus.

03 PRESSURE & OWNERSHIP

Who Owned the Gestures?

Apple's Moves

- Buys FingerWorks, 2005
- Jobs, Jan 2007: "boy, have we patented it"
- Patent 7,479,949 granted Jan 2009
- Cook, Jan 2009: "we will not stand for having our IP ripped off"

Microsoft's Moves

- Hires Buxton, Dec 2005
- Surface shown May 2007; in AT&T stores April 2008
- Windows 7 touch demo, May 2008
- Files patent on this study's method, Aug 2008 (inventors: Morris, Wobbrock, Wilson) — abandoned 2014

Documented

Apple and Microsoft have held a patent cross-licence since 1997. Both companies were filing aggressively around gesture interaction in 2007–2009. The study's patent application was filed on the same day its data collection concluded.

We Argue

This looks less like lawsuit avoidance and more like a race to *own conventions* — to establish whose gesture vocabulary becomes the standard before the ecosystem locks in. The abandoned patent suggests the value was strategic, not litigative.



04 THE METHOD

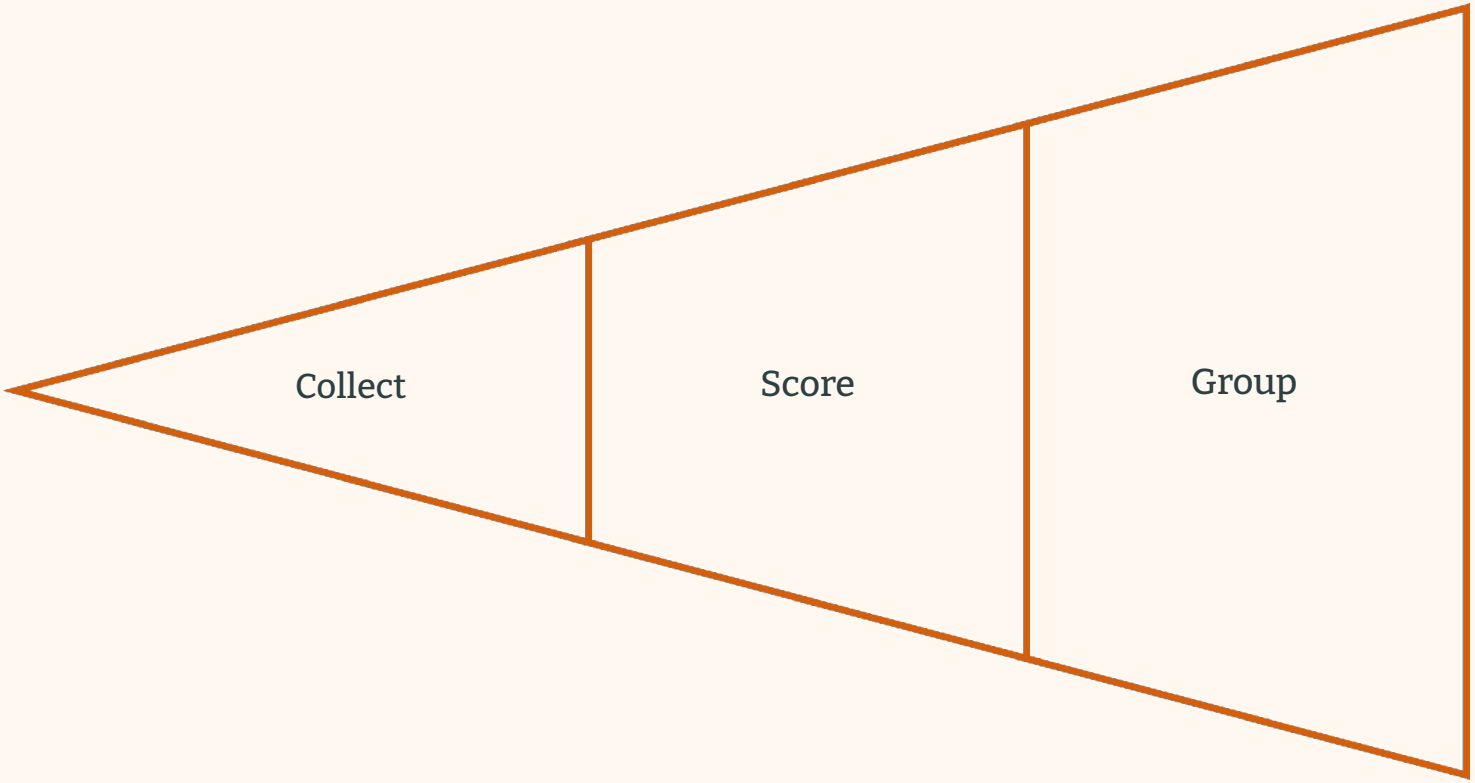
A Guessability Study: From Referent to Sign

Three methodological choices separated this study from prior gesture research.

1. A recorded voice named the command, for example "delete," and the table played a short animation of the effect, for example a block disappearing
2. The scene reset to its starting state and the table waited.
3. The participant performed the gesture they thought would have caused that effect, with one hand, thinking aloud, then again with two hands.
4. They said which of the two they preferred and rated each on 7-point scales.
5. The table logged the touches but did nothing in response.

04 THE METHOD

From 1,080 Gestures to a Final Set



Scoring Agreement (Eq. 1)

Agreement score **A** = mean over referents of Σ (group share)²

What the Funnel Cannot Do

- Gestures close/task switch
- Single gestures/sign which win
- Decision

05 HOW THE RESEARCH WAS CONDUCTED

How the study was run

Who and Where

- 20 paid volunteers, 9 women, mean age 43.2, all right-handed
- None had used a tabletop computer or an iPhone
- Microsoft Surface prototype, 24 × 18 inches, 1024 × 768 resolution

Procedure

- 27 referents shown in random order; authors rated complexity 1–5, mean 2.70
- Per referent: 1-hand gesture, then 2-hand gesture, think-aloud, preference, two 7-point ratings
- 1,080 gestures collected; 6 discarded; touch logs, video, and notes kept
- Baseline: each author made their own gestures *before* the study began

Revelations

Designers missed

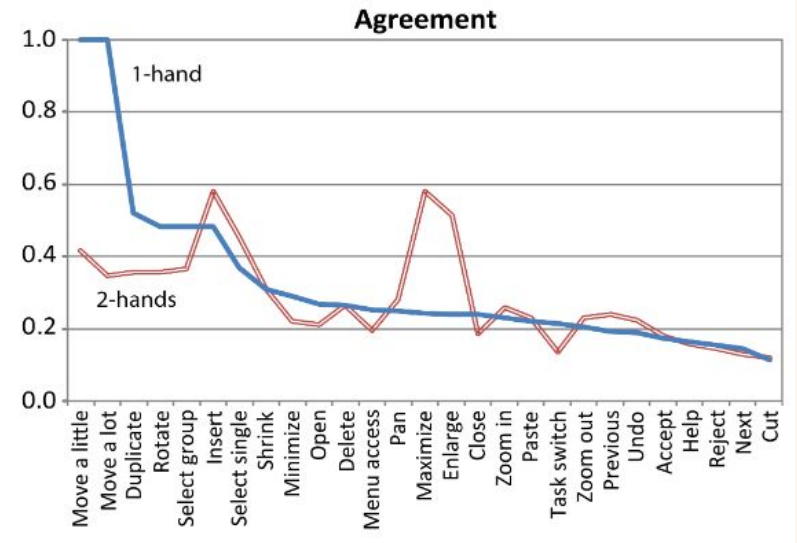
Each author matched only 43.5% of the user set; all three together matched 60.9%. 19.1% of each author's gestures no participant ever tried.

One hand wins

Preferred for 25 of 27 referents; tied for insert and maximize; two hands never preferred. Yet 17 of 48 final set gestures (35.4%) are two-handed driven by agreement, not preference.

Fingers barely matter

13 of 20 participants varied finger count freely; only 2 said count was meaningful. 1–3 fingers read as a "single point"; 4 was a "gray area"



06 SAMPLE & CULTURE

Who was in the room and who was not?

The Sample

20 adults, mean age 43.2, sd 15.6

No age range reported.

No ethnicity reported.

None had used a tabletop or iPhone

Culture Embedded in the Set

Several gestures in the final set are Western symbolic conventions: a drawn check mark for Accept, an X for Reject, a question mark for Help, a left-to-right line for Next. These symbols do not transfer cleanly across cultures:

- In Japan, ○ marks correct and ✓ can indicate wrong
- In Bulgaria, a head nod means *no*
- A thumbs-up is an insult in Iran

07 FINDINGS

The key findings which are worth mentioning

01

Desktop deeply rooted

~72% of gestures were mouse-like one-point touches or paths

02

One hand preferred

For 25 of 27 referents; two hands never the top pick

03

Agreement falls with complexity

Harder tasks produce more scattered gesture choices

04

Physical vs. symbolic

43.9% physical gestures; complex commands attracted metaphors and symbols

05

Finger count rarely matters

13 of 20 varied count; only 2 said it was meaningful

06

Nature of opposites

Never shown as pairs — participants spontaneously mirrored gestures

07

Beyond the surface

Off-screen drags, mid-air gestures, and imaginary widgets appeared

08

Popularity predicts ratings

More-popular gestures were rated better by their own makers

07 FINDINGS

More findings...

- **Thinking:** about 8 s before touching for simple commands, about 15 s for complex ones;
- **Doing:** about 2 to 3.4 s, and complexity made no difference to it (p. 1088)
- **Longer thinking went with lower "good match" ratings:** good gestures came to mind fast
- **We argue:** on a walk-up table the cost is in finding the gesture, not making it. What did they do because of this issue?
- **Speed is not ease:** gestures rated easy took about 3.4 s, rated difficult about 2.0s. Comfortable gestures were slow and smooth, awkward ones quick and precise
- **Caveat:** means only; 1,074 gestures from 20 people counted as independent
- **Bias:** There are so many biased elements inside this paper, where do i start?

08 TECHNICAL OBSTACLES

Technical limitations and implications

Hardware Constraints

- Display: 24" × 18" at 1024 × 768 approximately 43 pixels per inch. A current iPad Pro: 264 ppi.
- Sensing: a vision system "from beneath the table"; contacts logged as ovals with millisecond timestamps (p. 1086). Video cameras recorded sessions separately.
- No touch recognizer. No feedback to participants during gesture performance (p. 1085, p. 1090).

Implications for the Study

- Participants were told to touch the table yet some still gestured in the air and beyond the screen edge. The sensor could not capture these but the cameras.
- A physics engine [32] was proposed as a future recognizer; supported in principle by 43.9% physical gestures, but authors concede it "will probably not suffice" alone.
- The gestures in the study might not been recognizable at the time

09 LIMITATIONS

Limitations

The Authors say

- No revising earlier gestures once performed
- No application, organizational, or cultural context
- "Children, Eastern, or uneducated participants would behave differently"
- Set not validated; no recognizer built or tested

We Add

- n = 20: sufficient for strong consensus, insufficient for subgroup analysis
- Authors grouped and coded gestures themselves prior to the study
- The statistics count 1,074 gestures as if 1,074 different people had made them. Twenty did, so the p-values are too confident
- Desktop referents throughout; 43% of proposed gestures are not covered
- Microsoft authors, own prototype patent application unmentioned in the paper

10 CONNECTIONS

Prior readings and how it relates to this paper:

1

Research Contribution in Human-Computer Interaction - Wobbrock & Kientz, 2016

Methods need repeated validation

2

Usability Evaluation Considered harmful - Greenberg & Buxton, 2008

Early evaluation will prevent development

3

Evaluating User Interface Systems Research - Olsen, 2007

Shifting the design participation from designer to User refered as new design participants

4

The computer for the 21st centaury - Weiser, 1991

Illustrates boards likely inspired by Buxton in 1985

11 RESEARCH DIRECTIONS

Research Directions

The Authors' Agenda

- Validation of gestures explored in study
- Recognizability of gestures
- Implementation with vision-based gesture recognizer

What Actually Followed?

- The authors did do the validation
- Method was used in 216 studies

12 DISCUSSION

Tensions Worth Arguing About

Four key questions for the paper:

1 Should the most popular gesture win?

~72% of gestures were mouse-like; Does consensus from novices lock in old habits? Guessability vs. learnability vs. long-term efficiency.

2 Did the commands shape the answers?

Cut, paste, minimize, maximize, and task switch are desktop concepts. Who should choose the referents and what changes if they change?

3 Popularity or invention enjoyment?

Ratings are self-reports by gesture inventors. What would independent validation actually look like?

4 Is behavior without feedback the right target?

Real users adapt to system responses. Participants already gestured in the air and off-screen (p. 1090). What does elicitation miss by removing the loop?

Topics to discuss:

- **Floor or ceiling?** 19.1% of each author's gestures were never tried. Does elicitation find the most guessable gesture or the best one?
- **Symbolic gestures import writing.** The check, the X, and the question mark are Western literacy marks. The paper's own caveat names types of participants but the set encodes Western norms anyway.
- **Single users on a group device.** Minimize = "move to the bottom of the surface as defined by user's seating position" (Figure 4). A different direction for every person at the table.

Fun for us ...

1. **Testing the method:** While a clear task was to discuss this, i believe we should also experience this method first hand...
2. **Let's discuss more**